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1-D CNN-based timing synchronization leveraging post processing with perception assistance in UAV communications

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ABSTRACT

In unmanned aerial vehicle (UAV)-assisted orthogonal frequency division multiplexing (OFDM) systems, correct timing synchronization (TS) is vital for ensuring reliable data transmission and maintaining flight safety. However, in UAV-based wireless scenarios, the first-arriving propagation path is not necessarily the strongest one, and the presence of multipath components coupled with Doppler frequency shifts significantly degrades TS accuracy. Although a lightweight one-dimensional convolutional neural network (1-D CNN)-based TS method partially addresses these issues, its improvements are limited and it often requires network retraining in highly dynamic UAV-assisted communication scenarios. To overcome these limitations, a perception-assisted post-processing framework is proposed, which establishes a new paradigm for improving the TS correctness of 1-D CNN-based synchronizers without network retraining. This framework exploits echo signals at the ground base station (gBS) to estimate transmission paths, thereby extracting perceptual prior information. By employing the 1-D CNN, we first achieve an optimized timing metric (TM) and an initial time offset (TO). Subsequently, perception-assisted post-processing operations, including path matching, perception threshold design, and backward detection, are employed to refine the initial TO estimation. Simulation results demonstrate that the proposed method significantly reduces TS error probability relative to baseline methods while maintaining its robustness against parameter variations without network retraining.

1. Introduction

1.1. Background

Unmanned aerial vehicle (UAV) has been widely applied in wireless communication systems to improve spectral efficiency, area coverage, and channel capacity [1], such as UAV-assisted wireless sensor network (WSN) [2], UAV-assisted vehicle-to-everything (V2X) communication [3], UAV-assisted internet of thing (IoT) [4], and UAV-assisted orthogonal frequency division multiplexing (OFDM) systems [5]. By leveraging the potential for flexible deployment and high spectral efficiency, UAV-assisted OFDM systems have emerged as a promising solution for reliable communications in dynamic wireless scenarios [6]. However, the correctness improvement of the timing synchronization (TS) remains a critical challenge in such systems. In UAV-assisted OFDM systems, the correctness of TS directly affects the subsequent performance of signal processing [7], such as channel estimation [8] and symbol detection [9]. Incorrect TS seriously deteriorates the reliability of data transmission in UAV-assisted OFDM systems, thereby significantly reducing the flight safety and the mission execution efficiency [10]. However, in UAV-assisted OFDM systems, the first-arriving path is often not the strongest one in UAV-based wireless communication scenarios. Coupled with the inevitable multipath and Doppler interference, achieving correct TS is facing significant challenges [11, 12].

1.2. Related works

In UAV-assisted OFDM systems, the classical TS methods such as those based on cross-correlation [13–15], auto-correlation [16,17], and Park [18] can be adopted. However, when the first-arriving path is not the strongest path, the TS correctness of these classical TS methods [13–18] is significantly degraded [19]. In order to improve the TS correctness of these classical methods, a maximum likelihood (ML)-based TS method is proposed in [20,21]. However, this method involves an exhaustive search as well as likelihood function computation, which significantly increases computational complexity and processing

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latency, rendering it impractical for UAV-assisted communication scenarios with strict real-time constraints. To address this challenge, TS methods based on compressed sensing (CS) [7,22], extreme learning machine (ELM) [23–25], and one-dimensional convolutional neural network (1-D CNN) [26] have been successively proposed. In [22], the CS-based TS method iteratively constructs a sparse representation of the received reference signal to estimate the timing offset (TO). Although the CS-based TS method in [22] reduces the computational complexity of the ML-based TS method in [20] to a certain extent, its iterative reconstruction method still results in relatively high computational complexity. To address this issue, the ELM-based TS methods have been proposed in [23–25]. These methods employ feedforward neural networks rather than the operation of iteration processing, significantly reducing the computational complexity of the CS-based TS method in [22]. Furthermore, these methods leverage the advantage of the nonlinear mapping of ELM to extract the salient features of timing metric (TM), thereby improving the correctness of TS. However, the single-hidden-layer structure of ELM network inherently constrains its learning capacity. When encountering severe multipath and Doppler interference, these ELM-based methods [23–25] fail to reliably maintain their TS correctness. To address this issue, a 1-D CNN-based TS method is proposed in [26]. This method leverages convolutional layers to extract features of TM, thereby improving the TS correctness of these ELM-based TS methods. Nevertheless, the 1-D CNN-based TS method in [26] still suffers from low TS correctness and sometimes even network retraining in UAV-based wireless scenarios. Beyond the traditional and learning-based TS methods discussed above, recently developed integrated sensing and communication (ISAC) frameworks have introduced novel approaches, including multimodal semantic fusion [27] and multi-UAV beamforming with rate maximization [28]. Nevertheless, the TS challenge in high-mobility UAV communication scenarios remains unaddressed in these works.

1.3. Motivation and contributions

With the development of ISAC technology, perception-assisted communication has been widely applied [29–31], e.g., perception-assisted channel estimation [32], perception-assisted channel state information (CSI) feedback [33], and perception-assisted TS [10]. For perception-assisted TS, a low-complexity perception-assisted TS method is proposed in [10]. Based on the classical cross-correlation method [13], this method utilizes perceptual prior information to iteratively identify the range where the first-arriving path might be located, thereby improving the TS correctness. Nonetheless, the perception-assisted TS method in [10] requires highly accurate perceptual prior information, which hinders its practical application. Furthermore, only cross-correlation operations are employed rather than neural network-based approaches. Consequently, the nonlinear characteristics of wireless channels are not utilized, thus degrading the TS correctness in UAV-based wireless scenarios. Inspired by perception-assisted TS and the 1-D CNN approach, a novel perception-assisted post-processing scheme for 1-D CNN-based TS is proposed. In this scheme, the 1-D CNN extracts TM features while leveraging perception information to narrow the search to a minimal candidate set, enabling the precise identification of the first-arriving path. The goal of this method is to develop a perception assistance scheme that improves the TS correctness and robustness of the 1-D CNN-based TS method while avoiding retraining its neural network, thereby reducing the memory requirement for storing model parameters. The contributions of this paper are summarized as follows:

- We develop a perception-assisted post-processing method for the 1-D CNN-based TS in UAV-assisted OFDM systems. This method establishes a new paradigm that integrates perception information with features extracted from the 1-D CNN. This integration enhances the correctness and robustness of TS in highly dynamic UAV channels without requiring network retraining.
- We propose a perception-assisted path matching scheme. In this scheme, we construct a path matching vector based on perceptual prior information. By combining this vector with the TM vector, this scheme determine the potential region of the first-arriving path, serving as prior knowledge for subsequent threshold design and noise interference suppression. As far as we know, no existing TS methods have explored a perception-assisted path matching scheme in related literature.
- We develop a perception-assisted first-arriving path detection scheme. In this scheme, a perception-assisted threshold design and a backward detection strategy are developed to enhance the reliability of identifying the first-arriving path, thereby improving the TS correctness of the 1-D CNN-based TS. To the best of our knowledge, few works of TS leverage echo-based perceptual prior information to assist the design of detection threshold.

The rest of this paper is organized as follows: In Section 2, we introduce the perceptual signal model and the communication signal model, respectively. Subsequently, the proposed perception-assisted post-processing TS method is elaborated in Section 3. The results of the numerical simulations are presented in Section 4. Finally, the conclusion of this paper is provided in Section 5.

Notations: $(\cdot)^T$ and $(\cdot)^H$ stand for the transpose and conjugate transpose, respectively. $\mathbb{E}(\cdot)$ denotes the complex expectation operation, $X \sim U[a,b]$ denotes X follows a uniform distribution in $[a,b]$, and $\mathrm{rect}[\cdot]$ represents the rectangular function. The circularly symmetric complex Gaussian (CSCG) distribution with mean ω and variance σ^2 is denoted by $\mathcal{CN}(\omega,\sigma^2)$, and $\sim$ stands for "distributed as". $\lceil x \rceil$ rounds up the value of x, and $\lfloor x \rfloor$ rounds down the value of x. $\mathbb{C}^{M\times N}$ denotes the $M\times N$ dimensional complex matrix, and $\mathbb{R}^{M\times N}$ denotes the $M\times N$ dimensional real matrix. $\mathrm{Re}(x)$ and $\mathrm{Im}(x)$ denote the real and imaginary parts of the complex x, respectively.

2. System model

In this section, the system model is presented. In Section 2.1, we introduce the perceptual signal model. The communication signal model is provided in Section 2.2.

2.1. Perceptual signal model

In UAV-assisted OFDM systems, the transmitted signal at the ground base station (gBS) in the time-domain, denoted as $x(t)$, is given by [34]

$$x(t) = \sum_{m=0}^{M-1}\sum_{n=0}^{N-1} X(m,n) e^{j2\pi n\Delta f t} \cdot \mathrm{rect}\left[\frac{t-mT_s}{T_s}\right], \tag{1}$$

where M denotes the number of OFDM symbols, N represents the subcarrier number of each symbol, $X(m,n)$ is the transmission data on the nth subcarrier of the mth symbol with $m = 0,1,\ldots,M-1$ and $n = 0,1,\ldots,N-1$, Δf denotes subcarrier spacing, and T_s denotes the duration of the OFDM symbol including cyclic prefix (CP). From [35], $\Delta f = 1/T$ with T being the duration of an OFDM symbol without CP.

At the gBS, the received echo signal, denoted as $y_{\mathrm{echo}}(t)$, is given by [36]

$$y_{\mathrm{echo}}(t) = \sum_{p=1}^{P} \beta_p x(t-\tau_p) e^{j2\pi\upsilon_p t} + \sum_{\widetilde{p}=1}^{\widetilde{P}} \widetilde{\beta}_{\widetilde{p}} x(t-\widetilde{\tau}_{\widetilde{p}}) e^{j2\pi\widetilde{\upsilon}_{\widetilde{p}} t} + \widetilde{w}(t), \tag{2}$$

where P is the number of echo paths, τ_p, υ_p, and β_p respectively denote the path delay, Doppler frequency shift, and path gain of the pth echo path. $\widetilde{P}$ represents the number of false paths, $\widetilde{\tau}_{\widetilde{p}}$, $\widetilde{\upsilon}_{\widetilde{p}}$, and $\widetilde{\beta}_{\widetilde{p}}$ respectively denote the path delay, Doppler frequency shift, and path gain of the $\widetilde{p}$-th false path, and $\widetilde{w}(t)$ denotes additive white Gaussian noise (AWGN) with variance $\widetilde{\sigma}^2$. With the received $y_{\mathrm{echo}}(t)$, false path suppression method described in [32] is employed to suppress the majority of the false paths, while treating the remaining false paths that have not been

suppressed as noise. Consequently, the received echo signal on the nth subcarrier of the mth symbol after the false path suppression, denoted as $Y_{\text{echo}}(m,n)$, is given by

$$Y_{\text{echo}}(m,n) = X(m,n)\sum_{p=1}^{P}\beta_p e^{j2\pi\left(m\upsilon_p T_s - n\Delta f\tau_p\right)} + W(m,n), \tag{3}$$

where $W(m,n)$ denotes the AWGN term with variance σ^2.

In this paper, the received echo signal $\mathbf{Y}_{\text{echo}} \in \mathbb{C}^{M\times N}$ is utilized to extract perceptual prior information for assisting the post-processing of the 1-D CNN.

2.2. Communication signal model

At the UAV, the mth OFDM frequency-domain transmitted symbol $d_m(k)$ with $k = 0,1,\ldots,N-1$ is transformed into the time-domain via the inverse fast Fourier transform (IFFT) to form $\left\{s_m(n)\right\}_{n=0}^{N-1}$. According to [26], the nth sample of the mth symbol $s_m(n)$ is given by

$$s_m(n) = \frac{1}{\sqrt{N}}\sum_{k=0}^{N-1} d_m(k)\, e^{j\frac{2\pi nk}{N}}. \tag{4}$$

In this paper, the power of transmitted signal $s_m(n)$ is $P_t = \mathbb{E}\left\{\left|s_m(n)\right|^2\right\}$. After appending a CP of length N_g, the transmitted signal vector $\mathbf{s}_m \in \mathbb{C}^{(N+N_g)\times 1}$, i.e., $\mathbf{s}_m = \left(s_m(-N_g), s_m(-N_g+1), \ldots, s_m(N-1)\right)^T$, is generated.

At the gBS, the received nth sample of the mth OFDM symbol in the time-domain, denoted as $y_m(n)$, is given by [37]

$$y_m(n) = \sum_{l=1}^{L} h_{\tau_l} s_m\left(n-\tau_l-\widetilde{\tau}\right) e^{j2\pi\varepsilon\frac{n-\widetilde{\tau}}{N}} + w_m(n), \tag{5}$$

where L denotes the number of resolvable paths, h_{τ_l} and τ_l respectively represent the channel complex gain and path delay of the lth transmission path, $\widetilde{\tau}$ denotes unknown TO to be estimated, ε represents the normalized carrier frequency offset (CFO), and $w_m(n)$ is the AWGN term. From [24], $\widetilde{\tau} \sim U[0, N-1]$ and $w_m(n) \sim \mathcal{CN}\left(0,\sigma_n^2\right)$ with σ_n^2 being noise variance. To detect a complete reference signal in multipath scenarios, the N_w-length received signal vector $\mathbf{y} \in \mathbb{C}^{N_w\times 1}$ (i.e., $\mathbf{y} = \left[y(0), y(1), \ldots, y(N_w-1)\right]^T$) is observed with N_w satisfying $N_w = 2N + N_g$. Utilizing the received signal vector $\mathbf{y}$, we develop a perception-based post-processing method to enhance the accuracy of the 1-D CNN-based TS in [26].

3. Perception-assisted post-processing method for 1-D CNN-based TS

The perception-assisted post-processing method for 1-D CNN-based TS is elaborated in this section. In Section 3.1, the perception scheme is illustrated, from which the perceptual prior information is extracted. Then, in Section 3.2, the perception-assisted post-processing method is provided. The system model is depicted in Fig. 1.

3.1. Perception information extraction

To estimate the echo path parameters of the target UAV, the approximate ML method in [38] is referenced in this paper. Different from [38] estimating the echo parameters of a single path, this paper focuses on the multipath echo parameters of the target UAV. Based on the received symbol $Y_{\text{echo}}(m,n)$ (given by Eq. (3)) and the known transmitted symbol $X(m,n)$ at the gBS, the channel transfer function of echo signal, denoted as $H(m,n)$, is estimated by [32]

$$\begin{aligned} H(m,n) &= \frac{Y_{\text{echo}}(m,n)}{X(m,n)} \\ &= \sum_{p=1}^{P}\beta_p e^{j2\pi\left(m\upsilon_{\text{norm},p} - n\tau_{\text{norm},p}\right)} + \overline{W}(m,n), \end{aligned} \tag{6}$$

where $\upsilon_{\text{norm},p} = \upsilon_p T_s$ and $\tau_{\text{norm},p} = \Delta f\tau_p$ represent the normalized Doppler frequency and normalized delay of the pth echo path, respectively. $\overline{W}(m,n)$ represents the AWGN term with variance σ_w^2. Thus, the matrix $\mathbf{H} \in \mathbb{C}^{M\times N}$ is obtained with its (m,n)-th element being $H(m,n)$. In this paper, we employ $\mathbf{H}$ to estimate echo path parameters via its 2D periodogram. This procedure is outlined in **Algorithm 1**, and its detailed derivation is provided as follows.

3.1.1. 2D periodogram calculation

According to [38], in order to improve the frequency resolution, the matrix $\mathbf{H}$ undergoes zero-padding processing to form the matrix $\mathbf{H}_{\text{ZP}} \in \mathbb{C}^{M_0\times N_0}$, which is given by

$$\mathbf{H}_{\text{ZP}} = \begin{bmatrix} \mathbf{H} & \mathbf{0}_{M\times(N_0-N)} \\ \mathbf{0}_{(M_0-M)\times N} & \mathbf{0}_{(M_0-M)\times(N_0-N)} \end{bmatrix}. \tag{7}$$

where M_0 and N_0 denote the extended lengths of the Doppler and delay, respectively. By respectively denoting the oversampling factors along the Doppler and delay directions as L_υ and L_τ, we have $M_0 = L_\upsilon M$ and $N_0 = L_\tau N$. Consequently, a 2D periodogram is achieved by using the discrete symplectic Fourier transform (DSFT) to $\mathbf{H}_{\text{ZP}}$. By denoting this 2D periodogram as $\mathbf{Y}_{\text{Map}} \in \mathbb{C}^{M_0\times N_0}$, its (l,q)-th entry $Y(l,q)$ with $l = 0,1,\ldots,M_0-1$ and $q = 0,1,\ldots,N_0-1$ is given by

$$Y(l,q) = \frac{1}{M_0N_0}\sum_{m=0}^{M_0-1}\sum_{n=0}^{N_0-1} H_{\text{ZP}}(m,n) e^{-j2\pi\left(\frac{ml}{M_0}-\frac{nq}{N_0}\right)}. \tag{8}$$

3.1.2. Echo path parameter estimation

To estimate the echo path parameters of the target UAV, an iterative echo path parameter detection scheme is developed. In this scheme, the residual matrices of the 2D periodogram and channel transfer function are initialized as $\mathbf{Y}_{\text{Res}}^{(1)} = \mathbf{Y}_{\text{Map}}$ and $\mathbf{H}^{(1)} = \mathbf{H}$, respectively. Correspondingly, the flag of echo path existence, the number of echo paths, and the counter are initialized as $I_{\text{flag}} = 1$, $\widehat{P} = 0$, and $i = 1$, respectively. We iteratively detect the echo paths until $I_{\text{flag}} = 0$, thereby obtaining the number of echo paths $\widehat{P}$ and their corresponding path parameters $\left\{\widehat{\boldsymbol{\beta}}, \widehat{\boldsymbol{\upsilon}}, \widehat{\boldsymbol{\tau}}\right\}$. Here, $\widehat{\boldsymbol{\beta}}$, $\widehat{\boldsymbol{\upsilon}}$, and $\widehat{\boldsymbol{\tau}}$ represent the detected path gain, Doppler shift, and time delay of $\widehat{P}$ echo paths, respectively.

In the ith iteration, we search the index of the largest amplitude value in the residual matrix $\mathbf{Y}_{\text{Res}}^{(i)}$. By denoting the searched index as $\left(\widehat{l}_i, \widehat{q}_i\right)$ with $\widehat{l}_i \in \left\{0,1,\ldots,M_0-1\right\}$ and $\widehat{q}_i \in \left\{0,1,\ldots,N_0-1\right\}$, it is given by [32]

$$\left(\widehat{l}_i, \widehat{q}_i\right) = \arg\max_{l,q}\left\{\left|Y_{\text{Res}}^{(i)}(l,q)\right|^2\right\}. \tag{9}$$

According to $\left(\widehat{l}_i, \widehat{q}_i\right)$, the estimated values of the normalized Doppler shift and the normalized delay for the current strongest echo path, represented as $\widehat{\upsilon}_{\text{norm},i}$ and $\widehat{\tau}_{\text{norm},i}$, are given by

$$\begin{cases} \widehat{\upsilon}_{\text{norm},i} = \frac{\widehat{l}_i}{M_0} - \frac{1}{2} \\ \widehat{\tau}_{\text{norm},i} = \frac{\widehat{q}_i}{N_0} \end{cases}. \tag{10}$$

According to the residual $\mathbf{H}^{(i)}$, by substituting $\left(\widehat{\upsilon}_{\text{norm},i}, \widehat{\tau}_{\text{norm},i}\right)$ into Eq. (6) and treating the other paths as noise affecting the current strongest echo path, we have [38]

$$H^{(i)}(m,n) = \beta_i e^{j2\pi\left(m\widehat{\upsilon}_{\text{norm},i} - n\widehat{\tau}_{\text{norm},i}\right)} + \overline{W}_i(m,n), \tag{11}$$

where $\overline{W}_i(m,n)$ is the noise term that includes the contributions from other paths (not the current strongest echo path). From [38], the ML estimate $\widehat{\beta}_i$ is given by

$$\widehat{\beta}_i = \arg\min_{\beta_i}\left[\frac{1}{MN}\sum_{m=0}^{M-1}\sum_{n=0}^{N-1}\left|H^{(i)}(m,n) - \beta_i e^{j2\pi\left(m\widehat{\upsilon}_{\text{norm},i} - n\widehat{\tau}_{\text{norm},i}\right)}\right|^2\right]. \tag{12}$$

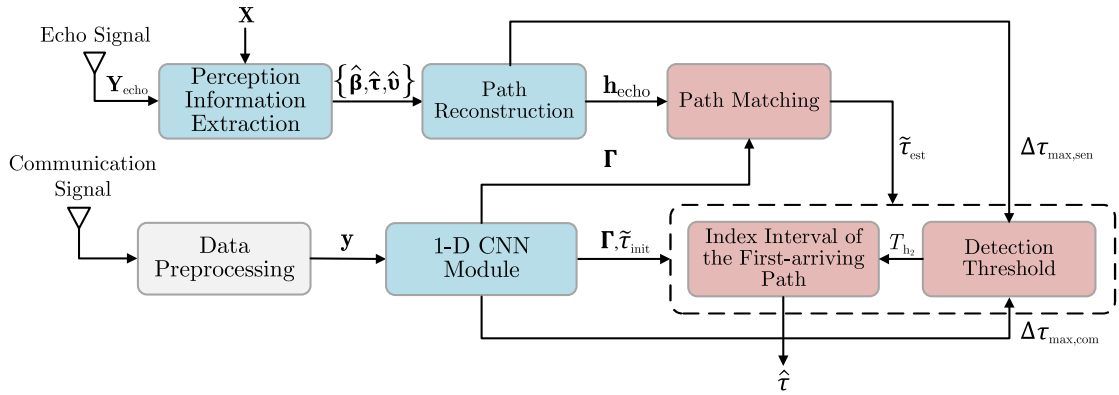

Fig. 1. system model.

From [38], $\hat{\beta}_i$ in Eq. (12) is approximately estimated by

$$\hat{\beta}_i = \frac{1}{MN}\sum_{m=0}^{M-1}\sum_{n=0}^{N-1} H^{(i)}(m,n)\, e^{-j2\pi\left(m\hat{\upsilon}_{\text{norm},i}-n\hat{\tau}_{\text{norm},i}\right)}. \tag{13}$$

To ensure that $\left(\hat{l}_i,\hat{q}_i\right)$ indexes a resolvable echo path, we design an adaptive detection threshold for resolvable paths. Denote this detection threshold as $T_{\text{h}_1}^{(i)}$, and according to [39], it is designed by

$$T_{\text{h}_1}^{(i)} = \frac{\alpha}{N_r}\sum_{(l,q)\in R}\left|Y_{\text{Res}}^{(i)}(l,q)\right|^2, \tag{14}$$

where R represents the set of reference units and N_r denotes the number of reference units in R. For a pre-defined false alarm probability P_{f_1}, the threshold factor α is derived as [40]

$$\alpha = N_r\left(P_{\text{f}_1}{}^{-1/N_r}-1\right). \tag{15}$$

Therefore, according to Eqs. (8) and (14), if $\left|Y_{\text{Res}}^{(i)}\left(\hat{l}_i,\hat{q}_i\right)\right|^2 < T_{\text{h}_1}^{(i)}$, then the index $\left(\hat{l}_i,\hat{q}_i\right)$ obtained in the ith iteration does not correspond to an echo path. Since $\left(\hat{l}_i,\hat{q}_i\right)$ is already the index of the element with the maximum amplitude value in the current periodogram, it is even less likely that there are other indexes corresponding to echo paths, and thus the iterative processing is terminated. If $\left|Y_{\text{Res}}^{(i)}\left(\hat{l}_i,\hat{q}_i\right)\right|^2 \geq T_{\text{h}_1}^{(i)}$, $\left(\hat{l}_i,\hat{q}_i\right)$ indicates the index of an echo path, thereby continuing the iterative processing.

If $\left|Y_{\text{Res}}^{(i)}\left(\hat{l}_i,\hat{q}_i\right)\right|^2 \geq T_{\text{h}_1}^{(i)}$, the residual measurement $\mathbf{H}^{(i+1)}$ is generated by subtracting the contribution of the strongest path to $\mathbf{H}^{(i)}$ based on the estimated $\left(\hat{\beta}_i,\hat{\upsilon}_{\text{norm},i},\hat{\tau}_{\text{norm},i}\right)$. Correspondingly, the contribution of the strongest echo path is eliminated from $\mathbf{Y}_{\text{Res}}^{(i)}$, thereby updating the residual spectrum as $\mathbf{Y}_{\text{Res}}^{(i+1)}$, i.e., [38]

$$\mathbf{Y}_{\text{Res}}^{(i+1)} = \mathbf{Y}_{\text{Res}}^{(i)} - \hat{\beta}_i\mathbf{C}^{(i)}, \tag{16}$$

where $\mathbf{C}^{(i)} \in \mathbb{C}^{M_0\times N_0}$ and its (l,q)-th element is given by

$$C^{(i)}(l,q) = \frac{1}{M_0N_0}\sum_{m=0}^{M-1}\sum_{n=0}^{N-1} e^{j2\pi\left(m\hat{\upsilon}_{\text{norm},i}-n\hat{\tau}_{\text{norm},i}\right)} e^{-j2\pi\left(\frac{ml}{M_0}-\frac{nq}{N_0}\right)}. \tag{17}$$

According to $\mathbf{Y}_{\text{Res}}^{(i+1)}$, we repeat the above steps from Eq. (9) to Eq. (17) to recursively estimate the parameters of the other echo paths until $\left|Y_{\text{Res}}\left(\hat{l},\hat{q}\right)\right|^2 < T_{\text{h}_1}$.

Finally, by using the above scheme, the complex gain vector $\hat{\boldsymbol{\beta}} = \left(\hat{\beta}_1,\hat{\beta}_2,\ldots,\hat{\beta}_{\hat{P}}\right)^T$, the normalized delay vector $\left(\hat{\tau}_{\text{norm},1},\hat{\tau}_{\text{norm},2},\ldots,\hat{\tau}_{\text{norm},\hat{P}}\right)^T$, and the normalized Doppler frequency shift vector $\left(\hat{\upsilon}_{\text{norm},1},\hat{\upsilon}_{\text{norm},2},\ldots,\hat{\upsilon}_{\text{norm},\hat{P}}\right)^T$ of the $\hat{P}$ resolvable echo paths of the target UAV are estimated. Thus, the estimated values of the delay and Doppler frequency for the pth path with $p = 1,\ldots,\hat{P}$, denoted as $\hat{\tau}_p$ and $\hat{\upsilon}_p$, respectively, are calculated by

$$\begin{cases} \hat{\tau}_p = \hat{\tau}_{\text{norm},p}/\Delta f \\ \hat{\upsilon}_p = \hat{\upsilon}_{\text{norm},p}/T_s \end{cases} \tag{18}$$

From Eqs. (13) and (18), the parameter set of $\hat{P}$ echo paths is constructed by $\left\{\left(\hat{\beta}_1,\hat{\tau}_1,\hat{\upsilon}_1\right),\ \left(\hat{\beta}_2,\hat{\tau}_2,\hat{\upsilon}_2\right),\ldots,\left(\hat{\beta}_{\hat{P}},\hat{\tau}_{\hat{P}},\hat{\upsilon}_{\hat{P}}\right)\right\}$. It should be noted that, $\{\hat{\tau}_1,\hat{\tau}_2,\ldots,\hat{\tau}_{\hat{P}}\}$ is arranged in ascending order of the delay of arrival at the gBS. Therefore, the parameters $\left\{\hat{\boldsymbol{\beta}},\hat{\boldsymbol{\tau}},\hat{\boldsymbol{\upsilon}}\right\}$ of the $\hat{P}$ resolvable echo paths are estimated.

However, due to the presence of false paths, the perceived echo paths may not be the echo paths of the target UAV. To address this issue, we employs the false path suppression method in [32] to suppress false paths. In this paper, we focus on the perception-assisted post-processing method for 1-D CNN-based TS, so the processing of false path suppression is not elaborated. Then, $\left\{\hat{\boldsymbol{\beta}},\hat{\boldsymbol{\tau}},\hat{\boldsymbol{\upsilon}}\right\}$ is viewed as the echo path parameters of the target UAV after false path suppression. Unlike the estimation of line of sight (LOS) path in [38], the parameters of resolvable echo paths, i.e., $\left\{\hat{\boldsymbol{\beta}},\hat{\boldsymbol{\tau}},\hat{\boldsymbol{\upsilon}}\right\}$, are estimated in this paper. In this paper, $\left\{\hat{\boldsymbol{\beta}},\hat{\boldsymbol{\tau}},\hat{\boldsymbol{\upsilon}}\right\}$ are regarded as perceptual prior information to assist the post-processing of the 1-D CNN-based TS, aiming to improve the TS correctness.

3.2. Perception-assisted post-processing of TS

Based on the perceptual prior information $\left\{\hat{\boldsymbol{\beta}},\hat{\boldsymbol{\tau}},\hat{\boldsymbol{\upsilon}}\right\}$ of the target UAV described in 3.1, this subsection details the perception-assisted post-processing of TS. The method of TO initial estimation based on 1-D CNN is first presented, followed by the proposed perception-assisted post-processing method.

3.2.1. Initial estimation of TO based on 1-D CNN

According to Eq. (5), by concatenating the real and imaginary parts of the complex-valued signal vector $\mathbf{y}$ received from the target UAV, a real-valued input $\tilde{\mathbf{y}} \in \mathbb{R}^{2N_w\times 1}$ of the 1-D CNN is constructed, i.e., [26]

$$\tilde{\mathbf{y}} = \left[\text{Re}\left(y(0)\right),\text{Im}\left(y(0)\right),\ldots,\text{Re}\left(y\left(N_w-1\right)\right),\text{Im}\left(y\left(N_w-1\right)\right)\right]^{\text{T}}. \tag{19}$$

According to Eq. (19), N_t training samples are collected to form the training set $\aleph$, which is given by

$$\aleph = \left\{\tilde{\mathbf{y}}_i,\mathbf{t}_i\right\}_{i=1}^{N_t}, \tag{20}$$

where $\tilde{\mathbf{y}}_i \in \mathbb{R}^{2N_w\times 1}$ and $\mathbf{t}_i \in \mathbb{R}^{N_u\times 1}$ (with $N_u = N_w - N_s + L$) represent the training data and label of the ith training sample, respectively. Unlike [26] that constructs training labels by using LOS prior information and the approximate midpoint of the ISI-free region, the proposed method directly employs the noise-free TM vector (i.e., the TM vector formed according to the received noise-free signals). By denoting this

Algorithm 1: Echo Path Parameter Estimation Algorithm

Input: Channel transfer function $\mathbf{H}$.
Output: Path parameters $\left\{\hat{\boldsymbol{\beta}}, \hat{\boldsymbol{\tau}}, \hat{\boldsymbol{\upsilon}}\right\}$ of $\hat{P}$ echo paths.

2D Periodogram Calculation:
Construct $\mathbf{H}_{\mathrm{ZP}}$ by using Eq. (7).
Utilize Eq. (8) to achieve the 2D periodogram $\mathbf{Y}_{\mathrm{Map}}$.
Echo Path Parameters Estimation:
Set the flag of existing echo paths, the counter, and the number of echo paths as $I_{\mathrm{flag}} = 1$, $i = 1$ and $\hat{P} = 0$. Initialize the residual matrices of 2D Periodogram and channel transfer function as $\mathbf{Y}_{\mathrm{Res}}^{(1)} = \mathbf{Y}_{\mathrm{Map}}$ and $\mathbf{H}^{(1)} = \mathbf{H}$, respectively.
while $I_{\mathrm{flag}} = 1$ **do**
  Estimate $\left(\hat{l}_i, \hat{q}_i\right)$, i.e., the entry index of the maximum amplitude in residual matrix $\mathbf{Y}_{\mathrm{Res}}^{(i)}$, by using Eq. (9);
  Calculate the normalized Doppler shift and delay $\left(\hat{\upsilon}_{\mathrm{norm},i}, \hat{\tau}_{\mathrm{norm},i}\right)$ via Eq. (10);
  Estimate the path gain $\hat{\beta}_i$ by using Eq. (12) and Eq. (13);
  Design the path detection threshold $T_{\mathrm{h}_1}^{(i)}$ according to Eq. (14);
  if $\left|Y_{\mathrm{Res}}^{(i)}\left(\hat{l}_i, \hat{q}_i\right)\right|^2 \geq T_{\mathrm{h}_1}^{(i)}$ **then**
    Generate the residual measurement $\mathbf{H}^{(i+1)}$;
    Update residual matrix $\mathbf{Y}_{\mathrm{Res}}^{(i+1)}$ by using Eq. (16);
    $i = i + 1$;
    $\hat{P} = \hat{P} + 1$;
  else
    $I_{\mathrm{flag}} = 0$;
    break;
  end
end
if $\hat{P} \neq 0$ **then**
  Utilize Eq. (18) to achieve $\left(\hat{\tau}_1, \hat{\tau}_2, \cdots \hat{\tau}_{\hat{P}}\right)$ and $\left(\hat{\upsilon}_1, \hat{\upsilon}_2, \cdots, \hat{\upsilon}_{\hat{P}}\right)$, thereby forming the parameter set of $\hat{P}$ resolvable echo paths, i.e., $\left\{\hat{\boldsymbol{\beta}}, \hat{\boldsymbol{\tau}}, \hat{\boldsymbol{\upsilon}}\right\}$.
end

received signal as $\bar{\mathbf{y}} \in \mathbb{C}^{N_w \times 1}$, the jth element of the ith training label $t_{i,j}$ is given by [13]

$$t_{i,j} = \left|\bar{\mathbf{s}}^H \bar{\mathbf{y}}_{j:j+N_s-1}\right|, j = 0, \ldots, N_w - N_s + L - 1, \tag{21}$$

where $\bar{\mathbf{s}} \in \mathbb{C}^{N_s \times 1}$ represents the known training sequence. By using the training set $\aleph$ to iteratively train the 1-D CNN, the network parameter set, denoted as Θ, is expressed as

$$\Theta = f_{\mathrm{Train}}(\aleph), \tag{22}$$

where $f_{\mathrm{Train}}(\aleph)$ represents the mapping function for obtaining the network parameters by using the training set $\aleph$. The mean squared error (MSE) is employed as the loss function, and the Adam optimizer is utilized for optimization (with the optimizer parameters $\alpha = 0.001, \beta_1 = 0.9, \beta_2 = 0.999$) [26].

For online running, the received complex-valued signal is transformed to the real-valued $\tilde{\mathbf{y}}$ by using Eq. (19). Based on the trained Θ, an optimized TM vector $\boldsymbol{\Gamma} \in \mathbb{R}^{N_u \times 1}$ is formed by employing the 1-D CNN, which is represented as

$$\boldsymbol{\Gamma} = f_{\mathrm{CNN}}(\Theta, \tilde{\mathbf{y}}), \tag{23}$$

where $f_{\mathrm{CNN}}(\Theta, \tilde{\mathbf{y}})$ represents the mapping function of the 1-D CNN timing synchronizer. By expressing the generated TM $\boldsymbol{\Gamma}$ as $\boldsymbol{\Gamma} = \left(\Gamma_0, \Gamma_1, \ldots \Gamma_{N_u-1}\right)^T$, the initial estimate of TO is determined by

$$\tilde{\tau}_{\mathrm{init}} = \underset{0 \leq j \leq N_u - 1}{\arg\max} \left\{\left|\Gamma_j\right|\right\}. \tag{24}$$

However, the estimated $\tilde{\tau}_{\mathrm{init}}$ identifies the strongest path, which may not necessarily be the first-arriving one. Consequently, in Section 3.2.2, we will use $\tilde{\tau}_{\mathrm{init}}$ to determine the index interval of the first-arriving path in the TM vector.

3.2.2. Perception-assisted post-processing method

In this subsection, a path matching scheme is first developed, followed by the schemes of perception-assisted threshold design and backward detection.

Path matching. To determine the index interval of the first-arriving path in the TM vector and reduce the degradation due to the network noise, we perform a coarse path matching between the TM vector and the perceptual prior information. This processing provides the necessary prior information for identifying the first-arriving path (including threshold design and backward detection).

Based on the perceptual prior information $\left\{\hat{\boldsymbol{\beta}}, \hat{\boldsymbol{\tau}}, \hat{\boldsymbol{\upsilon}}\right\}$, the composite channel response of the pth gBS-UAV-gBS echo path with $p = 1, 2, \ldots, \hat{P}$ is expressed as

$$\tilde{h}_p = \hat{\beta}_p \delta\left(t - \hat{\tau}_p\right) e^{j2\pi\hat{\upsilon}_p}. \tag{25}$$

Relative to the path echo delay (i.e., the gBS-UAV-gBS path delay), the communication path delay (i.e., the UAV-gBS path delay) is usually only its half. Therefore, based on the sampling interval T/N, the delay set Ω_{delay}, which is formed by the $\left\{\frac{\hat{\tau}_1}{2}, \frac{\hat{\tau}_2}{2}, \ldots, \frac{\hat{\tau}_{\hat{P}}}{2}\right\}$, is given by

$$\Omega_{\mathrm{delay}} = \left\{\left\lceil\frac{N\hat{\tau}_1}{2T}\right\rceil, \left\lceil\frac{N\hat{\tau}_2}{2T}\right\rceil, \ldots, \left\lceil\frac{N\hat{\tau}_{\hat{P}}}{2T}\right\rceil\right\}. \tag{26}$$

Therefore, the number of sampling points in $\left[\frac{\hat{\tau}_1}{2}, \frac{\hat{\tau}_{\hat{P}}}{2}\right]$, denoted as N_{echo}, is given by

$$N_{\mathrm{echo}} = \left\lceil\frac{N\hat{\tau}_{\hat{P}}}{2T}\right\rceil - \left\lceil\frac{N\hat{\tau}_1}{2T}\right\rceil + 1. \tag{27}$$

By constructing the path matching vector as $\mathbf{h}_{\mathrm{echo}} = \left(\breve{h}_1, \breve{h}_2, \ldots, \breve{h}_{N_{\mathrm{echo}}}\right)^T \in \mathbb{R}^{N_{\mathrm{echo}} \times 1}$, its nth entry $\breve{h}_n$ with $n = 1, 2, \ldots, N_{\mathrm{echo}}$ satisfies

$$\breve{h}_n = \begin{cases} \left|\tilde{h}_n\right|, & \text{if } n \in \Omega_{\mathrm{delay}} \\ 0, & \text{otherwise} \end{cases} \tag{28}$$

By performing sliding correlation according to $\mathbf{h}_{\mathrm{echo}}$ and $\boldsymbol{\Gamma}$ [13], a correlation output vector, denoted as $\mathbf{R} \in \mathbb{R}^{(N_u - N_{\mathrm{echo}}+1) \times 1}$, is achieved by

$$\mathbf{R} = \mathbf{H}_{\mathrm{echo}}^H \boldsymbol{\Gamma}, \tag{29}$$

where $\mathbf{H}_{\mathrm{echo}} \in \mathbb{R}^{N_u \times (N_u - N_{\mathrm{echo}}+1)}$ is the cyclic matrix formed by the cyclic shift of $\mathbf{h}_{\mathrm{echo}}$, i.e.

$$\mathbf{H}_{\mathrm{echo}} = \begin{bmatrix} \breve{h}_1 & 0 & \cdots \\ \vdots & \breve{h}_1 & \ddots \\ \breve{h}_{N_{\mathrm{echo}}} & \vdots & \ddots \\ 0 & \breve{h}_{N_{\mathrm{echo}}} & \ddots \\ \vdots & 0 & \ddots \\ \vdots & \vdots & \ddots \end{bmatrix}. \tag{30}$$

By expressing $\mathbf{R}$ as $\mathbf{R} = \left[R_0, R_1, \ldots, R_{N_u - N_{\mathrm{echo}}}\right]^T$, the time index with the strongest correlation, denoted as $\tilde{\tau}_{\mathrm{est}}$, is estimated as

$$\tilde{\tau}_{\mathrm{est}} = \underset{0 \leq j \leq N_u - N_{\mathrm{echo}}}{\arg\max} \left|R_j\right|. \tag{31}$$

According to the estimated $\tilde{\tau}_{\mathrm{est}}$ and $\tilde{\tau}_{\mathrm{init}}$, the index of the first-arriving path lies within the interval $\mathcal{T}$, which is expressed as

$$\mathcal{T} = \left[\tilde{\tau}_{\mathrm{est}} - \Delta\tau, \tilde{\tau}_{\mathrm{init}}\right], \tag{32}$$

where $\Delta\tau$ is employed to act as a expansion factor. Due to the unavoidable inaccuracy of path matching, we employ an expansion factor to increase the interval range, thereby improving the interval correctness. According to the index interval $\mathcal{T}$, the corresponding TM interval can be derived for backtracking detection.

Perception-assisted threshold design. Based on the path matching information $\widetilde{\tau}_{\text{est}}$, the initial estimation of TO $\widetilde{\tau}_{\text{init}}$, the TM vector $\boldsymbol{\Gamma}$, and the maximum multipath delay of the UAV-gBS wireless channels, the detection threshold for identifying the first-arriving path is designed. Here, we first derive the maximum multipath delay. Then, an index set of non-path TM elements is constructed.

According to Eq. (5), when there is no signal received at the gBS (i.e., $y_m(n)$ contains only noise $w_m(n)$), the TM vector $\widetilde{\boldsymbol{\Gamma}}$ (generated by the 1-D CNN) is employed to statistically determine its mean $\mu_{\widetilde{\Gamma}}$ and variance $\sigma^2_{\widetilde{\Gamma}}$. Based on the central limit theorem [41], $\widetilde{\Gamma}_j \sim \mathcal{CN}\left(\mu_{\widetilde{\Gamma}}, \sigma^2_{\widetilde{\Gamma}}\right)$, and the detection threshold $T_{\text{h}_{\text{pre}}}$ is derived as [19]

$$\int_{T_{\text{h}_{\text{pre}}}}^{\xi} \frac{1}{\sqrt{2\pi\sigma^2_{\widetilde{\Gamma}}}} \exp\left[-\frac{\left(x-\mu_{\widetilde{\Gamma}}\right)^2}{2\sigma^2_{\widetilde{\Gamma}}}\right] = P_{\text{F,pre}}, \tag{33}$$

where $P_{\text{F,pre}}$ represents the pre-defined false alarm probability, and ξ denotes the maximum value of $\left\{\widetilde{\Gamma}_j\right\}$ obtained through long-term statistics. Based on the detection threshold $T_{\text{h}_{\text{pre}}}$, the indices of TM elements exceeding the threshold are collected, forming the set $\left\{\widetilde{\tau}_1, \widetilde{\tau}_2, \ldots, \widetilde{\tau}_{\widetilde{L}}\right\}$ with $\widetilde{L}$ denoting the number of these TM elements. Consequently, the maximum multipath delay from the TM vector $\widetilde{\boldsymbol{\Gamma}}$ is estimated by

$$\Delta\tau_{\text{max,com}} = \widetilde{\tau}_{\widetilde{L}} - \widetilde{\tau}_1. \tag{34}$$

According to the perception information $\left(\widehat{\tau}_1, \widehat{\tau}_2, \ldots \widehat{\tau}_{\hat{P}}\right)$, the perceptual maximum multipath delay, denoted as $\Delta\tau_{\text{max,sen}}$, is given by

$$\Delta\tau_{\text{max,sen}} = \left\lceil \left(\frac{\widehat{\tau}_{\hat{P}}}{2} - \frac{\widehat{\tau}_1}{2}\right)\frac{N}{T} \right\rceil. \tag{35}$$

Consequently, the maximum multipath delay of the resolvable paths does not exceed the maximum value between $\Delta\tau_{\text{max,sen}}$ and $\Delta\tau_{\text{max,com}}$. By denoting an upper bound of the maximum multipath delay of the resolvable paths as $\Delta\tau_{\text{max}}$, we have

$$\Delta\tau_{\text{max}} = \max\left\{\Delta\tau_{\text{max,sen}}, \Delta\tau_{\text{max,com}}\right\}. \tag{36}$$

Based on the interval $\mathcal{T}$, the index set of non-path TM elements, denoted as $\boldsymbol{\Omega}$, is given by

$$\boldsymbol{\Omega} = \left[0, \widetilde{\tau}_{\text{est}} - \Delta\tau - 1\right] \cup \left[\widetilde{\tau}_{\text{init}} + \Delta\tau_{\text{max}} + 1, N_u - 1\right]. \tag{37}$$

Consequently, based on Γ_j with $j \in \boldsymbol{\Omega}$, the detection threshold, denoted as T_{h_2}, is designed by

$$\int_{T_{\text{h}_2}}^{\Gamma_{\widetilde{\tau}_{\text{est}}}} \frac{1}{\sqrt{2\pi\widehat{\sigma}^2}} \exp\left[-\frac{\left(x-\overline{x}\right)^2}{2\widehat{\sigma}^2}\right] = P_{\text{f}_2}, \tag{38}$$

where $\overline{x}$ and $\widehat{\sigma}^2$ respectively represent the mean and variance of Γ_j, and P_{f_2} is the pre-defined false alarm probability.

Backward detection based on perceptual threshold. Based on Eq. (32) and the detection threshold T_{h_2} (obtained from Eq. (38)), the refined estimation of TO is given by

$$\widetilde{\tau} = \underset{j \in \left[\widetilde{\tau}_{\text{est}} - \Delta\tau, \widetilde{\tau}_{\text{init}}\right]}{\text{argfirst}} \left\{\left|\Gamma_j\right| \geq T_{\text{h}_2}\right\}, \tag{39}$$

where $\text{argfirst}\{\cdot\}$ represents the operation of finding the first index that satisfies $\left|\Gamma_j\right| \geq T_{\text{h}_2}$.

For OFDM systems, a correct TS must ensure that the start of the DFT window falls within ISI-free region. Similar to [10], this paper exploits the perceptual prior information to locate the synchronization point at the midpoint of the ISI-free region. Combining the refined estimation $\widetilde{\tau}$ obtained from Eq. (39), the estimated midpoint of the ISI-free region, denoted as $\widetilde{\tau}_{\text{mid}}$, is given by

$$\widetilde{\tau}_{\text{mid}} = \widetilde{\tau} - \left\lfloor \frac{N_g - \Delta\tau_{\text{max,sen}} - 1}{2} \right\rfloor. \tag{40}$$

In this paper, $\widetilde{\tau}_{\text{mid}}$ is regarded as the correct TS point for OFDM within the ISI-free region [10].

4. Numerical simulation

In this section, we validate the effectiveness and robustness of the proposed method. Definitions and basic parameters involved in simulations are first given in Section 4.1. In Section 4.2, we analyze the computational complexity. Following this, the effectiveness of the proposed method is presented in Section 4.3. Finally, we validate the robustness of the proposed method in Section 4.4.

4.1. Parameter setting

The basic parameters involved in the simulations are given as follows. $M = 64$, $N = 128$, $N_g = 32$, and the oversampling factors are $L_\upsilon = L_\tau = 4$. The pre-defined false alarm probabilities are set as $P_{\text{f}_1} = 10^{-6}$ and $P_{\text{F,pre}} = P_{\text{f}_2} = 10^{-5}$, respectively. The UAV velocity is $\upsilon = 100$km/h, $N_w = 2N + N_g = 288$, and the TO is considered to be uniformly distributed over the interval $[0, N-1]$, i.e., $\tau \sim U[0, N-1]$ [24]. The carrier frequencies for both the perception signal (i.e., gBS-UAV-gBS signal) and the communication signal (i.e, UAV-gBS signal) are set as 2.1 GHz [42]. The subcarrier spacing is 15 kHz. The training sequence for uplink synchronization employs the Zadoff–Chu (ZC) sequence [43]. The 1-D CNN is trained in exponential decay channel scenarios, where the number of training samples, the number of multipaths, and the channel decay factor are set as $N_t = 1 \times 10^4$, $L = 28$, and $\eta = 0.3$, respectively. During online running, the multipath number and channel decay factor of wireless channels are set as $L = 23$ and $\eta = \ln 10/(L-1) \approx 0.1$ [44]. It is worth noting that due to the effect of the radar cross section (RCS), there is a difference in path gain between the communication and echo channels. Therefore, in the simulation, we randomly generate echo channel gains to reflect this difference. According to [44,45], the TS error probability is employed as the metric of TS correctness in this paper, which is defined as the probability of the estimated TO $\widetilde{\tau}_{\text{mid}}$ falling outside of the ISI-free region. Unless otherwise specified, the subsequent simulations in this paper adopt the parameters provided in this section. For the convenience of expression, the abbreviations of the baseline methods and the proposed methods used in these simulations are summarized in Table 1.

4.2. Computational complexity analysis

According to [10], we use complex multiplications (CMs) as the metric to estimate the computational complexity. The computational complexity and running time of TS methods are elaborated in Table 2. From [22], the "CS" requires calculating the inner product of all candidate elements with the residuals and solving the least squares problem in each iteration of reconstruction. Thus, its CMs are $N_g N_s N_c + \sum_{l=1}^{N_g}\left(3lN_c + l^2 N_c + l^3\right)$, where N_g denotes the number of iterations. In contrast, the "CNN" in [26] designs a lightweight network composed of three 1-D CNN layers and a FCNN layer to reduce computational complexity, yielding its CMs are $2.5N_c^2 + N_g N_c + 7N_c$. In addition, the "DL" in [46] uses the real and imaginary parts of the received signal in parallel as input. These components are processed through two CNN layers followed by a fully connected layer with, employing 32 convolutional kernels. This results in the CMs of $8N_w\left(N_w + 2210\right)$ [46]. The "LC-Sen" in [10] employs cross-correlation to obtain the TM vector and then iteratively estimates the corresponding index interval of first-arriving path according to the multipath number (derived from the perceptual echo signals), leading to CMs of $\left(N_w - N_s + N_{\text{echo}}\right)N_s$ with N_{echo} denoting

Table 1
The abbreviations of the methods used in this paper.

Methods	Descriptions
CS	Compressive sensing-based TS method in [22].
CNN	Lightweight 1-D CNN-based TS method in [26].
DL	Deep learning-based TS method in [46].
LC-Sen	Low complexity sensing-aided TS method in [10].
Prop(without P-P)	Proposed TS method without perception-assisted post-processing.
Prop	The proposed method in this paper.

Table 2
The analysis of computational complexity.

Methods	Computational complexity	Case example	Running time (ms)
CS	$N_g N_s N_c + \sum_{l=1}^{N_s} \left(3lN_c + l^2 N_c + l^3\right)$	3,017,984	115.0184
CNN	$2.5N_c^2 + N_g N_c + 7N_c$	70,240	28.5919
DL	$8N_w \left(N_w + 2210\right)$	5,755,392	38.5399
LC-Sen	$\left(N_w - N_s + N_{\text{echo}}\right) N_s$	23,424	7.2171
Prop	$2.5N_u^2 - N_g N_u + 3N_u + 0.25N_{\text{echo}} \left(N_u - N_{\text{echo}} + 1\right)$	83,863	54.3234

the number of iterations. Relative to "CNN", the "Prop" reduces one CNN layer. Thus the CMs for network processing in "Prop" are $2.5N_u^2 - N_g N_u + 3N_u$. Additionally, the perception-assisted post-processing of "Prop" requires additional CMs of $0.25N_{\text{echo}} \left(N_u - N_{\text{echo}} + 1\right)$ according to the sliding correlation operation as given in Eq. (29). In this processing, each correlation involves N_{echo} real multiplications and the correlation slides $(N_u - N_{\text{echo}})$ times. By considering the correlation with the operation of sliding, we need $N_{\text{echo}} \left(N_u - N_{\text{echo}} + 1\right)$ correlation operations in total, leading to $N_{\text{echo}} \left(N_u - N_{\text{echo}} + 1\right)$ real multiplications (approximately equal to $0.25N_{\text{echo}} \left(N_u - N_{\text{echo}} + 1\right)$ CMs). According to that one complex multiplication is approximately equivalent to four real multiplications, the additional CMs are $0.25N_{\text{echo}} \left(N_u - N_{\text{echo}} + 1\right)$ CMs. Consequently, the total CMs of "Prop" are $2.5N_u^2 - N_g N_u + 3N_u + 0.25N_{\text{echo}} \left(N_u - N_{\text{echo}} + 1\right)$.

For the case where $N_w = 288$, $N_s = 128$, $N_g = 32$, $N_u = N_w - N_s + L = 188$, $N_c = N_w - N_s = 160$, and $N_{\text{echo}} = 23$, Table 2 presents an example of computational complexity. From Table 2, the CMs of "CS", "CNN", "DL", "LC-Sen", and "Prop" are 3,017,984, 70,240, 5,755,392, 23,424, and 83,863, respectively. In the provided example, the CMs of "CS" and "DL" are significantly larger than that of "Prop". This indicates that "Prop" substantially reduces the computational complexity of "CS" and "DL" by leveraging a lightweight neural network. As shown in Table 2, "Prop" achieves a lower running time than "CS" and "DL", rendering it more suitable for UAV-based communication scenarios. In comparison with "CNN" and "LC-Sen", "Prop" has slightly higher CMs and running time. This is because "Prop" employs the TM vector as the training label to learn all path features, causing its label length to be larger than that of "CNN". Additionally, the utilization of perception-assisted post-processing method in "Prop" introduces additional CMs, thus leading to a slightly higher running time than those of "CNN" and "LC-Sen". Despite this, "Prop" achieves a significant improvement in TS correctness compared to "CNN" and "LC-Sen", which will be illuminated in Section 4.3.

4.3. Effectiveness analysis

To analyze the effectiveness of the proposed TS method, Fig. 2 depicts the performance of the TS error probability. As depicted in Fig. 2, for each given signal-to-noise ratio (SNR), the TS error probability of "Prop" is consistently lower than those of "CS", "CNN", "DL", and "LC-Sen". For the case where SNR is 8 dB, the TS error probabilities of all baseline methods are greater than 1.48×10^{-2}, whereas the TS error probability of "Prop" is only 2.44×10^{-3}. This indicates that compared to the baseline methods, "Prop" significantly reduces their TS error probabilities. From Fig. 2, "CS" encounters the maximum TS error probability due to the inaccurate number of iterations. For "CS", it is difficult to accurately obtain the iteration number. Thus, similar

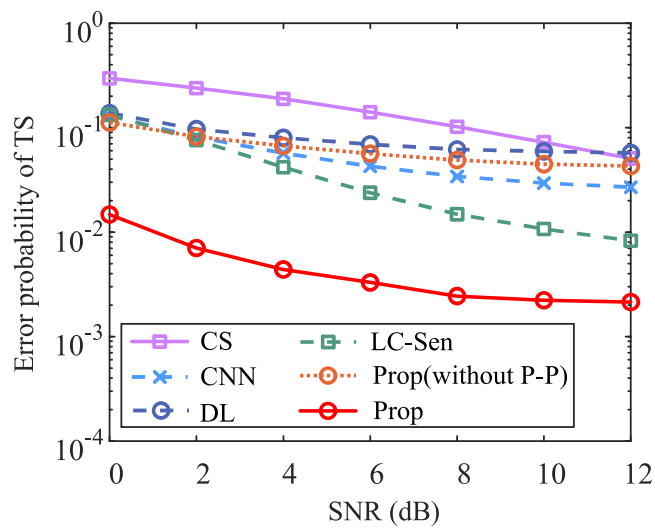

Fig. 2. Effectiveness analysis.

to [10], we adopt its CP length. For "CNN", "DL", and "LC-Sen", the Doppler shift caused by UAV movement interferes with TS, thereby increasing their TS error probabilities. In contrast, "Prop" employs perception assistance and threshold technique (as shown in Section 3.2), which effectively reduce the TS error probability. As shown in Fig. 2, all curves tend to flatten when the SNR exceeds 10 dB. This is because the detection threshold has stabilized in relatively high SNR region, making the TS error probability bound by the fundamental constraint of the ISI-free region length and the inherent accuracy of the TS algorithm.

Additionally, the proposed post-processing method based on the detection threshold is also validated to be effective in reducing the TS error probability. For example, for the case where SNR = 6 dB, the TS error probability of "Prop (without P-P)" is approximately 5.63×10^{-2}. This is because, without the operation of post-processing, the TS point obtained by the CNN corresponds to index of the strongest path, which significantly deteriorates the TS correctness. Relative to "Prop (without P-P)", the TS error probability of "Prop" is reduced to 3.3×10^{-3}. This improvement is attributed to the design of the perception threshold, which detects the resolvable paths that arrive before the strongest path and reduces the impact of spurious peaks generated by the CNN on the detection of the first-arriving path.

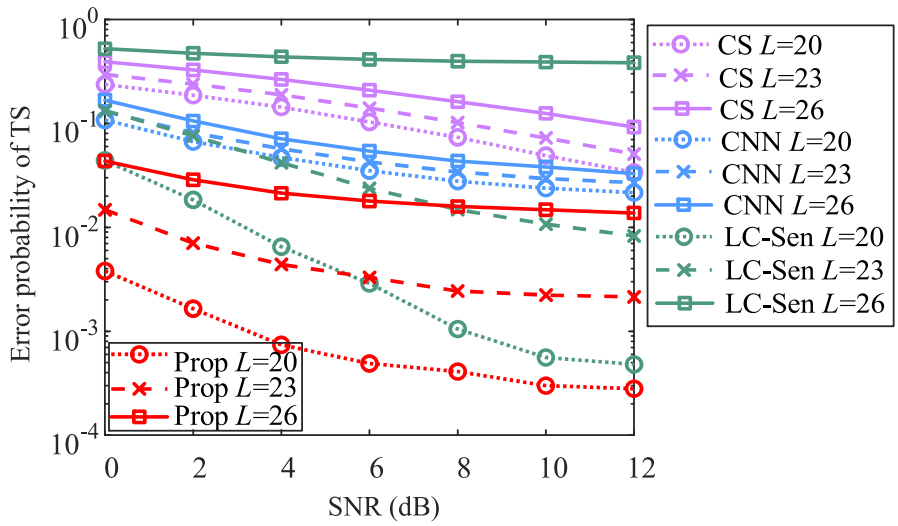

Fig. 3. Impact of different values of L on error probability of TS.

4.4. Robustness analysis

To demonstrate the robustness of the proposed method, we analyze its TS error probability performance against the impact of parameter variations in this subsection. It is noteworthy that, except for the variations of the impact parameters discussed in this section, the other parameters remain the same as those given in Section 4.1.

4.4.1. Impact of L

To illustrate the impact of the number of multipath on the TS error probability, Fig. 3 depicts the TS error probabilities for various values of L, where $L = 20$, $L = 23$, and $L = 26$ are considered. As shown in Fig. 3, as L increases, the TS error probabilities of all given methods increase for a specific SNR. This is because an increased L not only causes more severe multipath interference but also simultaneously reduces the length of the ISI-free region. Despite these challenges, compared to "CS", "CNN", and "LC-Sen", "Prop" consistently achieves a lower TS error probability for each given SNR and L. For example, for the case where SNR = 6 dB and $L = 20$, the TS error probability of "Prop" is 4.9×10^{-4}, whereas the TS error probabilities of the baseline methods are no less than 3.3×10^{-3}. This demonstrates that, even against the variations of L, the proposed method significantly enhances the performance of TS error probability relative to the baseline methods.

4.4.2. Impact of P_{f_2}

To analyze the impact of the false alarm probability P_{f_2}, Fig. 4 illustrates the TS error probabilities for different values of P_{f_2}, i.e., $P_{f_2} = 10^{-4}$, $P_{f_2} = 10^{-5}$, and $P_{f_2} = 10^{-6}$. As shown in Fig. 4, for each given SNR and P_{f_2}, the TS error probability of "Prop" remains nearly unchanged. This indicates that the threshold employed in the proposed method is robust against the variations of P_{f_2}. In contrast, the baseline methods do not utilize threshold detection, and thus their TS error probabilities are independent of the P_{f_2}. Nevertheless, even against the variations of P_{f_2}, the proposed method still demonstrates superior performance of TS error probability compared to the baseline methods.

4.4.3. Impact of η

Fig. 5 plots the TS error probabilities for different exponential decay factors η, where $\eta = 0.05$, $\eta = 0.1$ and $\eta = 0.2$ are considered. As depicted in Fig. 5, for a given SNR, the TS error probabilities of "Prop" and "CNN" decrease with the increase of η. This is because a higher η makes the power attenuation difference of each path more significant, which is more conductive to estimate the first-arriving path in "Prop" and "CNN". In contrast, the TS error probabilities of "CS" and "LC-Sen" increase with the increase of η. This is because these methods rely on the estimation of full path gains to identify the first-arriving path, and increasing η makes it difficult to estimate the path with a smaller gain. Nevertheless, for a given SNR and η, "Prop" demonstrates superior advantage in TS error probability performance compared to the baseline methods. For instance, when $\eta = 0.2$, starting from SNR = 2 dB, the error probability of "Prop" is no more than 0.6×10^{-3}, while the error probabilities of the baseline methods are no less than 1.75×10^{-3}. Overall, against the impact of exponential decay factor η, the proposed method reduces the TS error probability compared to the baseline methods.

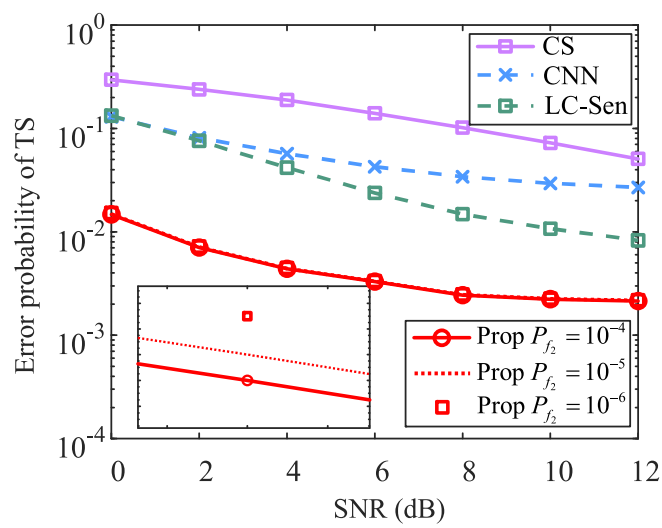

Fig. 4. Impact of different values of P_{f_2} on error probability of TS.

4.4.4. Impact of SNR_{echo}

To illustrate the impact of the echo SNR (i.e., SNR_{echo}), Fig. 6 depicts the TS error probabilities of "Prop" under different SNR_{echo} conditions, where SNR_{echo} = 10 dB, SNR_{echo} = 15 dB and SNR_{echo} = 20 dB are considered. As observed in Fig. 6, for each given SNR, the TS error probability of "Prop" slightly decreases as the SNR_{echo} increases from 10 dB to 15 dB. This is because a higher SNR_{echo} makes more accurate perception information, thereby reducing the TS error probability. Further increasing the SNR_{echo} to 20 dB, the TS error probability of "Prop"

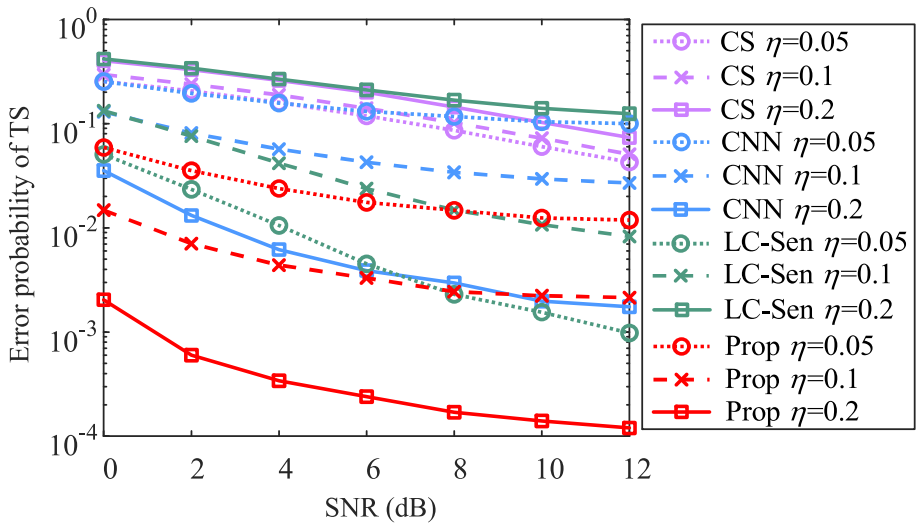

Fig. 5. Impact of different values of η on error probability of TS.

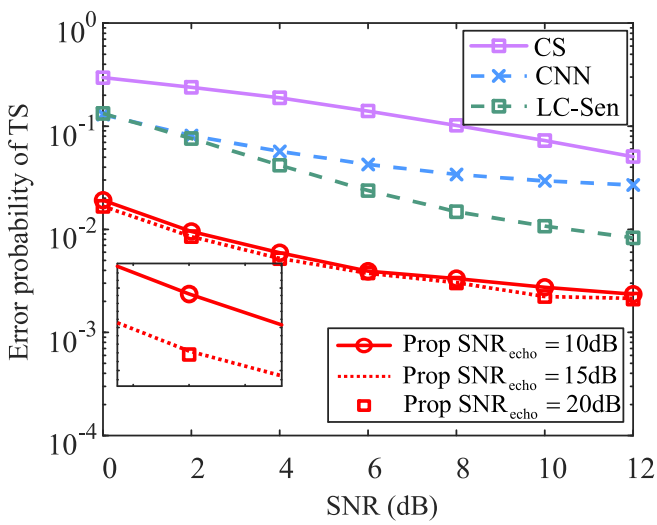

Fig. 6. Impact of different values of $\mathrm{SNR}_{\mathrm{echo}}$ on error probability of TS.

is almost the same as that of $\mathrm{SNR}_{\mathrm{echo}}$ = 15 dB. Due to $\mathrm{SNR}_{\mathrm{echo}}$ = 15 dB is large enough, further increasing it to 20 dB has a limited improvement on the TS correctness. Nevertheless, for each given SNR and $\mathrm{SNR}_{\mathrm{echo}}$, the TS error probability of "Prop" remains lower than those of baseline methods. For example, when SNR = 8 dB, the TS error probabilities of "Prop" with the values of $\mathrm{SNR}_{\mathrm{echo}}$ being 10 dB, 15 dB, and 20 dB are 3.33×10^{-3}, 3.04×10^{-3}, and 3.01×10^{-3}, respectively. In contrast, the TS error probabilities of "CS", "CNN", and "LC-Sen" are 1.02×10^{-1}, 4.264×10^{-2}, and 2.377×10^{-2}, respectively. Overall, for the given value of $\mathrm{SNR}_{\mathrm{echo}}$, the proposed method still improves the TS error probability performance of the baseline methods.

4.4.5. Impact of v

To demonstrate the robustness of the proposed method against high-mobility conditions, this subsection investigates the impact of UAV velocity v. A higher velocity introduces more severe Doppler spread, which is a primary impairment in dynamic UAV channels. Fig. 7 plots the TS error probabilities of "Prop" for different UAV velocities, where v = 100 km/h, v = 150 km/h, and v = 200 km/h are considered. From Fig. 7, it can be observed that for a given SNR, as the value of v increases, the TS error probabilities of "Prop", "CS", and "LC-Sen" also increase. This is because a higher velocity results in a larger Doppler shift, which degrades the TS error probability performance. However, the "CNN" remains a comparable TS error probability with the variations of v for a specific SNR. Even so, for a specific SNR and v, the TS error probability of "Prop" is lower than those of "CNN", "CS", and "LC-Sen". For example, when v = 200 km/h and SNR = 6 dB, the error probability of "Prop" is 5.04×10^{-3}, while the error probabilities of the baseline methods are all greater than 3.45×10^{-2}. Overall, against the variations of UAV velocity, "Prop" still improves the TS error probability performance of the baseline methods. To evaluate the impact of Doppler spread on TS performance, the TS error probability is depicted in Fig. 8, where the Doppler spreads in $[0.5f_{\mathrm{d}}, f_{\mathrm{d}}]$, $[0, f_{\mathrm{d}}]$, and $[-f_{\mathrm{d}}, f_{\mathrm{d}}]$ are considered with f_{d} denoting the maximum Doppler shift at a UAV speed of 100 km/h. As shown in Fig. 8, the TS error probability increases with the increase of Doppler spread for each given SNR. Even so, the proposed method still demonstrates superior TS performance for given Doppler spreads.

4.4.6. Impact of channel scenarios

Based on the parameter settings in Section 4.1, the 1-D CNN is trained in the scenarios of exponential decay channels. To validate the performance under high-mobility conditions with complex impairments, Fig. 9 illustrates the impact of different channel scenarios on TS error probabilities, where the tapped-delay-line (TDL)-D and TDL-E channel scenarios are considered. These scenarios are more consistent with UAV communication scenarios according to 3GPP TR 38.901 [47] and effectively represent scenarios with fast fading. As observed in Fig. 9, for given channel scenarios, the TS error probabilities of "Prop" is lower than those of baseline methods. For instance, in the TDL-D channel scenarios, when SNR = 4 dB, the TS error probability of "Prop" is 3.3×10^{-4}, whereas the TS error probabilities of the other methods are all greater than 1.04×10^{-3}. In the TDL-E channel scenarios, for a given SNR, "Prop" also achieves a smaller TS error probability compared to other methods. Overall, compared to the baseline methods, the proposed method significantly reduces the TS error probability in different channel scenarios, demonstrating its strong robustness against typical fast fading of high-mobility scenarios.

5. Conclusion

In this paper, the perception-assisted post-processing method for 1-D CNN-based TS method in UAV-assisted OFDM systems has been investigated. In this method, the 1-D CNN is employed to generate an optimized TM vector, thereby achieving an initial estimate of TO. With the optimized TM vector, we match the perceptual prior information to

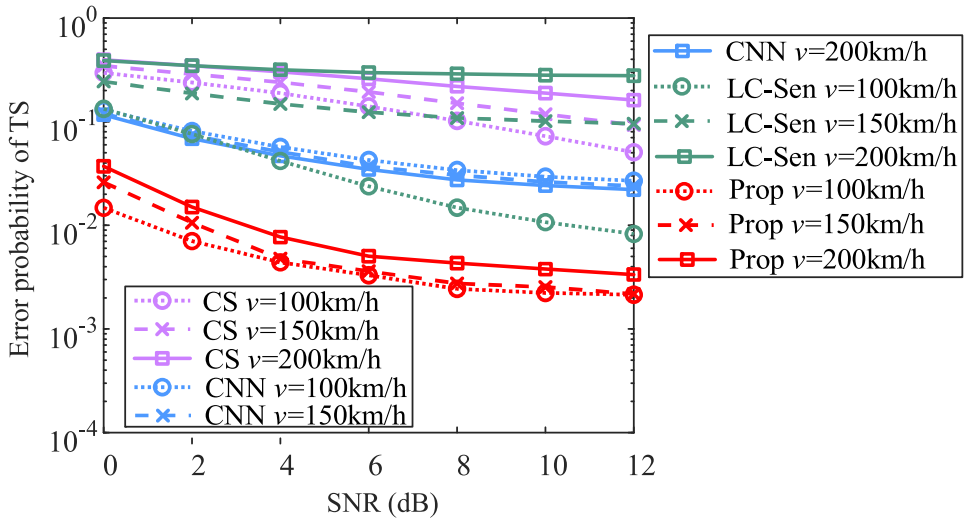

Fig. 7. Impact of different values of v on error probability of TS.

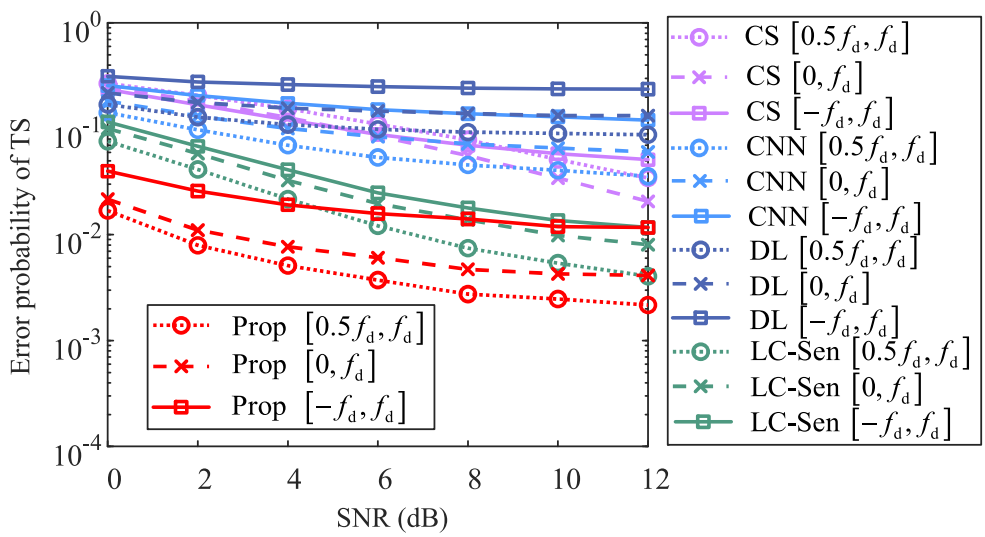

Fig. 8. Impact of different doppler spread on error probability of TS.

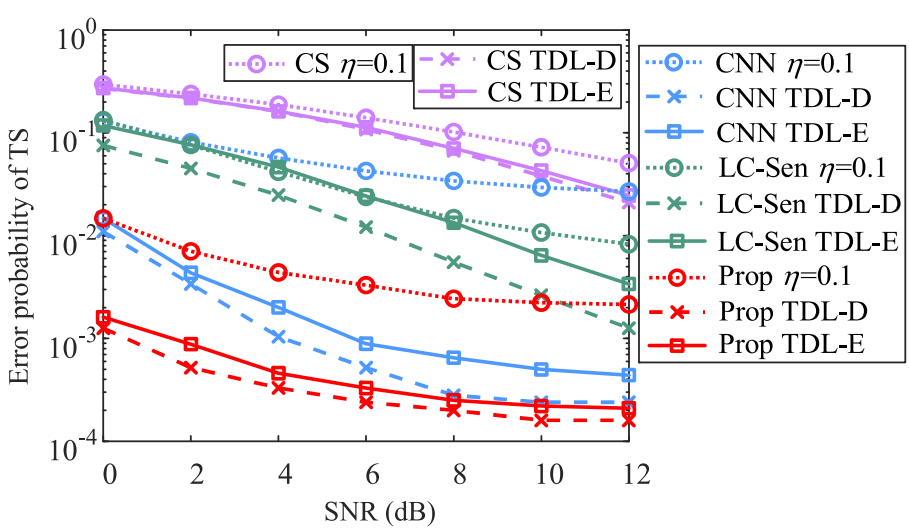

Fig. 9. Impact of different channel scenario on error probability of TS.

determine the TM interval in which the first-arriving path may exist. On this basis, according to the path matching information, the initial estimate of the TO, the TM, and the maximum multipath delay of the UAV-gBS wireless channels, the detection threshold for identifying the first-arriving path is designed. Consequently, the refined estimation of TO is achieved by backtracking the first-arriving path according to the TM interval and the detection threshold. The proposed method significantly narrows down the search range of the first-arriving path compared to the baseline methods by fusing the path features of perception and communication signals. With the perception-assisted adaptive detection threshold, the correctness of detecting the first-arriving path is improved. Moreover, the 1-D CNN employs the TM vector as the training label. This enables the trained network to match the perception information in UAV-based wireless scenarios, thereby facilitating the improvement of the TS correctness in different wireless channel environments. The simulation results demonstrate that, compared to the baseline methods, the proposed method significantly reduces the TS error probability. Against parameter variations, the proposed method maintains superior performance of TS error probability compared to the baseline methods even without retraining the network.

CRediT authorship contribution statement

Chaojin Qing: Conceptualization, Formal analysis, Funding acquisition, Methodology, Project administration, Supervision, Writing – review & editing. **Qiuting Hu:** Conceptualization, Investigation, Methodology, Software, Validation, Visualization, Writing – original draft, Writing – review & editing. **Yuxin Huang:** Conceptualization, Investigation, Methodology, Supervision. **Qian Zhao:** Formal analysis, Investigation, Methodology, Supervision. **Pengfei Du:** Data curation, Funding acquisition, Resources, Validation. **Mian Ye:** Data curation, Funding acquisition, Resources, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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